INCH-POUND
A-A-59652
31 August 2001
SUPERSEDING
FF-B-187/4
10 September 1993

COMMERCIAL ITEM DESCRIPTION

BEARING, ROLLER, TAPERED, SINGLE ROW, STEEP ANGLE, FLANGED CUP (TYPE TSSF)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes government acquisition requirements for flanged cup, steep angle, single row, tapered (type TSSF), roller bearings for general purpose use.

2. CLASSIFICATION. The roller bearings shall be of one type (TSSF) and classified by the size codes listed in table I. The column headings in table I refer to bearing characteristics defined in figure 1.

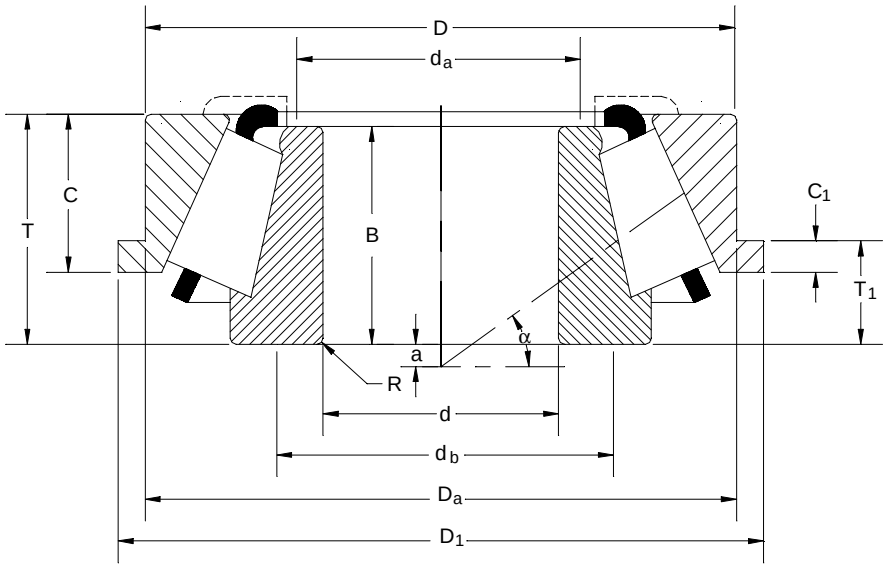

FIGURE 1. Bearing configuration.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3110

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TABLE I. Size codes and dimensions.*

Size code	Part 1/ number	d	D	T	T1	B	C	R 2/	D1	C1	d b	d a	D a	K factor	Basic dynamic load ratings (lb.)	a 3/
		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
001	21075-21212-B	0.7500	2.1250	0.8750	0.4062	0.8598	0.6250	0.06	2.2772	0.1562	1.24	1.04	2.05	0.99	7500	-0.23
002	23100-23256-B	1.0000	2.5625	0.8750	0.4063	0.8450	0.6250	0.06	2.7147	0.1563	1.54	1.36	2.48	0.80	88100	-0.09
003	41126-41286-B	1.1250	2.8593	0.9688	0.5000	0.9550	0.6875	0.06	3.0741	0.2187	1.63	1.45	2.76	0.97	11200	-0.16
004	43125-43312-B	1.2500	3.1250	1.0000	0.5313	0.9478	0.6875	0.06	3.3390	0.2188	1.73	1.63	3.03	0.08	12500	-0.08
005	43131-43312-B	1.3125	3.1250	1.0000	0.5313	0.9478	0.6875	0.14	3.3390	0.2188	2.01	1.65	3.03	0.08	12500	-0.08
006	43132-43312-B	1.3125	3.1250	1.0000	0.5313	0.9478	0.6875	0.08	3.3390	0.2188	1.89	1.66	3.03	0.08	12500	-0.08
007	44143-44348-B	1.4375	3.4843	1.0000	0.5312	0.9330	0.6875	0.09	3.6983	0.2187	2.13	1.97	3.39	0.09	13600	0.09
008	44162-44348-B	1.6250	3.4843	1.0000	0.5312	0.9330	0.6875	0.09	3.6983	0.2187	2.24	2.00	3.39	0.09	13600	0.09
009	53176-53387-B	1.7500	3.8750	1.2188	0.6563	1.1142	0.8125	0.05	4.1210	0.2500	2.32	2.08	3.62	0.79	16200	-0.01
010	55209C-55437-B	2.0000	4.3750	1.1875	0.5938	1.0594	0.8125	0.14	4.5938	0.2188	2.80	2.54	4.21	0.03	20600	0.30
011	55206-55437-B	2.0625	4.3750	1.1875	0.5938	1.0594	0.8125	0.14	4.5938	0.2188	2.83	2.52	4.21	0.03	20600	0.28
012	55206C-55437-B	2.0625	4.3750	1.1875	0.5938	1.0594	0.8125	0.14	4.5938	0.2188	2.83	2.54	4.21	0.03	20600	0.30
013	66225-66462-B	2.2500	4.6250	1.3125	0.6250	1.2500	0.9375	0.14	4.8750	0.2500	2.99	2.71	4.45	0.93	24000	-0.01
014	HM911249-HM911210-B	2.4375	5.1250	1.4375	0.7500	1.3125	0.9375	0.14	5.3750	0.2500	3.35	2.93	4.87	0.71	26900	0.21
015	9380-9321-B	3.0000	6.7500	1.9375	1.0000	1.8125	1.2500	0.14	7.0620	0.3125	4.13	3.87	6.46	0.76	49100	0.17
016	9386H-9321-B	3.3125	6.7500	1.9375	1.0000	1.8125	1.2500	0.14	7.0620	0.3125	4.37	3.87	6.46	0.76	49100	0.17
017	98400-98788-B	4.0000	7.8740	2.0772	1.0772	1.9375	1.3750	0.14	8.2500	0.3750	5.04	4.76	7.40	0.92	65200	0.05
018	37425-37625-B	4.2500	6.2500	0.9063	0.4688	0.8440	0.6250	0.14	6.4336	0.1875	4.80	4.53	6.02	0.54	18900	0.54
019	37431-37625-B	4.3125	6.2500	0.9063	0.4688	0.8440	0.6250	0.14	6.4336	0.1875	4.84	4.57	6.02	0.54	18900	0.54

1/ Part numbers are for reference only. Part numbers reflect the cup and cone numbers used by industry and the American Bearing Manufacturers Association (ABMA).
2/ These maximum fillet radii shall be cleared by the bearing corners.
3/ Minus value indicates load center inside cone backface.
* Column headings in the above table refer to bearing characteristics defined in figure 1.

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3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing dimensions (and dynamic load ratings, see 3.4) shall conform to the requirements specified in table I for each of the bearing part numbers. The listed dimensions conform to the requirements specified in American National Standards Institute /American Bearing Manufacturers Association (ANSI/ABMA) Standard 19.2, "Tapered Roller Bearings - Radial Inch Design" (DoD adopted). The bearing size shall be specified in the acquisition order (see 7.4(b)). For any unlisted bearing size codes, the associated dimensional and dynamic load rating requirements should also be specified in the acquisition order.

3.2 Materials.

3.2.1 Cones (inner rings), cups (outer rings), and rollers. The bearing cones, cups, and rollers shall be made of case carburized or through-hardened steel produced in accordance with the American Society for Testing and Materials (ASTM) A 295, "Standard Specification for High-Carbon Anti-Friction Bearing Steel" (DoD adopted), ASTM A 534, "Standard Specification for Carburizing Steels for Anti-Friction Bearings" (DoD adopted), or ASTM A 535, "Standard Specification for Special-Quality Ball and Roller Bearing Steel" (DoD adopted). The steel shall show a fine fracture grain size in accordance with ASTM E 112, "Standard Test Methods for Determining Average Grain Size" (DoD adopted). Material hardness shall be no less than 58 HRC and no more than 64 HRC as defined in ASTM E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials" (DoD adopted).

3.2.2 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The cages shall be made from carbon steel (one piece stamped). Materials shall operate from -65 to 230 °F.

3.3 Tolerance class. The tolerance limits for bearings shall conform to tolerance class 4 as tabulated in ABMA Standard 19.2. Allowable tolerances for bearing components and assembled bearings are listed in tables II through VI.

TABLE II. Cone bore tolerance.

Cone bore (d)			
Size range B		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	6.0000	10	0

Note: Allowable tolerances are in 0.0001 inch.

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TABLE III. Cup diameter tolerance.

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE IV. Standout tolerance.

Standout (T1)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	4.0000	80	0
4.0000	6.0000	140	100

Note: Allowable tolerances are in 0.0001 inch.

TABLE V. Cup flange tolerance.

Cup flange OD (D1)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	12.0000	20	0

Note: Allowable tolerances are in 0.0001 inch

TABLE VI. Assembled bearing tolerance.

Assembled bearing maximum radial runout		
Cup OD (D)		Tolerance
Over	Incl.	
0	24.0	20

Note: Allowable tolerances are in 0.0001 inch.

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3.4 Dynamic load rating. The bearing dynamic load rating shall conform to the requirements specified in table I for each bearing size code. The listed ratings conform to the requirements specified in ABMA Standard 11, "Load Ratings and Fatigue Life for Roller Bearings".

3.5 Effective load center. Dimension (a) in figure 1 locates a point on the cone axis, which is the center of pressure of all resisting forces set up by the rollers. All moments should be calculated from this point when determining bearing loading and shaft stresses. A plus value of (a) indicates that the center is outside the cone backface.

3.6 Lubrication and preservation. The bearings shall be furnished without lubrication and shall be preserved for packaging in accordance with MIL-DTL-197, "Packaging of Bearings, Antifriction Associated Parts and Subassemblies". If a preservative compound is required for packing or storage, the compound name shall be specified in the acquisition order (see 7.4(c)).

3.7 Steep angle. A steep angle bearing has a contact angle between 22 and 31 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Unless otherwise indicated in the solicitation and resulting contract, the foreign acquisition restrictions in Section 225.7019 of the Defense Federal Acquisition Regulation Supplement (DFARS) apply to products described by this CID.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the bearings supplied shall be preserved, packed, and marked as specified in the acquisition order (see 7.4(d)).

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7. NOTES

7.1 Sources of documents.

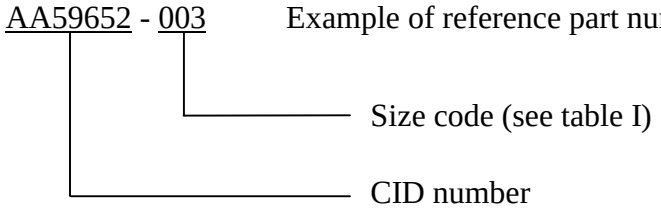
7.1.1 FAR and DFARS. Copies of the FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far/. Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dp/dars/dfars.html.

7.1.2 Military details. Copies of military details may be obtained from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies of military specifications and standards may be obtained from http://astimage.daps.dla.mil/quicksearch/.

7.1.3 ABMA standards. Copies of ABMA standards may be obtained from the American Bearing Manufacturers Association, 1200 19th Street, Suite 300, Washington, DC 20036-2401.

7.1.4 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



AA59652 - 003 indicates: bearing bore 1.12500 inches; outside diameter 2.8593 inches; width 0.9688 inches.

7.3 Sources of supply. The manufacturers and/or suppliers listed below are known to supply products that meet the salient characteristics requirements of this document. Competition is not limited to the listed firms.

KOYO Corporation of U.S.A.
Westlake, OH 44145

FAG Bearings Corporation
Danbury, CT 06813

NTN Bearing Corporation
Mount Prospect, IL 60056

SKF Industries
Kulpsville, PA 19443

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7.4 Ordering data. Acquisition documents shall specify the following information:

a. CID document number, revision, and CID PIN
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1)
c. Preservative compound, if required (see 3.6)
d. Preservation, packing, and marking requirements (see 6.1)

7.5 Cross-reference information. Table VII relates the original specification slant sheets to the replacement CIDs.

TABLE VII. Cross-reference table.

FF-B-187B	Replacement CID	ABMA type
Specification sheet 1	A-A-59649	TS
Specification sheet 2	A-A-59650	TSF
Specification sheet 3	A-A-59651	TSS
Specification sheet 4	A-A-59652	TSSF
Specification sheet 5	A-A-59653	TDI & TDIS
Specification sheet 6	A-A-59654	TDO
Specification sheet 7	A-A-59655	TDOS
Specification sheet 8	A-A-59656	TNA (normal angle)
Specification sheet 9	A-A-59657	TNAS (steep angle)
Specification sheet 10	A-A-59658	TNASW
Specification sheet 11	A-A-59659	TNASWE

7.6 Subject term (key word) listing.

bore
cone
cup
load
width

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MILITARY INTERESTS:

Custodians:
Army - AR
Navy - YD
Air Force - 99

Reviewers:
Navy - MC, OS
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - GS4

(Project 3110-1240)

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